

Maharashtra Public Service Commission Assessment

Subject Name :	Assistant Professor Biochemistry Group A Higher and Technical Education Department
Duration :	60
Number of Questions :	100
Total Marks :	200
Display Marks:	Yes
Maximum Instruction Time :	0
Is Section Default? :	No

Question Number : 1 Question Id : 4410092155711 Question Type : MCQ

Which sequence type most readily forms Z-DNA?

Options :

1. Alternating adenine and guanine
2. Only adenine and thymine
3. Alternating cytosine and guanine
4. Only guanine

Question Number : 2 Question Id : 4410092155714 Question Type : MCQ

Which type of vitamins can be extracted from foods using aqueous solvents?

Options :

1. Fat-soluble vitamins
2. Isoprenoid vitamins
3. Water-soluble vitamins
4. Hormone precursor vitamins

Question Number : 3 Question Id : 4410092155713 Question Type : MCQ

Which of the following protein complex is essential for microRNA-mediated gene silencing in eukaryotic cells?

Options :

1. Nucleosome core particle
2. Spliceosome complex
3. RNA-induced silencing complex (RISC)
4. DNA polymerase holoenzyme

Question Number : 4 Question Id : 4410092155716 Question Type : MCQ

Which structural component of vitamin K undergoes a redox cycle during the activation of prothrombin?

Options :

1. Side chain
2. Phosphate group
3. Sugar moiety
4. Aromatic ring

Question Number : 5 Question Id : 4410092155718 Question Type : MCQ

Which vitamin serves as the precursor of the coenzyme pyridoxal phosphate (PLP)?

Options :

1. Thiamine (vitamin B1)
2. Niacin (vitamin B3)
3. Pyridoxine (vitamin B6)
4. Riboflavin (vitamin B2)

Question Number : 6 Question Id : 4410092155707 Question Type : MCQ

Which of the following best describes a 'helix-turn-helix' motif in proteins?

Options :

1. It consists of alternating beta sheets and alpha helices connected by loops.
2. It consists of two alpha helices joined by a short sequence of amino acids forming a turn.
3. It consists of a single continuous alpha helix forming a rigid rod-like structure.
4. It consists of three alpha helices connected by extended loop regions.

Question Number : 7 Question Id : 4410092155709 Question Type : MCQ

How does the phosphodiester bond orientation influence the directionality of nucleic acid synthesis?

Options :

1. It enforces 5' to 3' synthesis by connecting the 3' hydroxyl of one nucleotide to the 5' phosphate of the next
2. It enables 3' to 5' synthesis by joining the 5' hydroxyl to the 3' phosphate
3. It is determined primarily by the sequence of nitrogenous bases
4. It allows for random addition of nucleotides due to alternating 2' and 4' positions

Question Number : 8 Question Id : 4410092155712 Question Type : MCQ

Which organelles contain transfer RNAs that are smaller than cytosolic tRNAs?

Options :

1. Nucleus and mitochondria
2. Cytosol and nucleus
3. Mitochondria and chloroplasts
4. Cytosol and ribosomes

Question Number : 9 Question Id : 4410092155715 Question Type : MCQ

Which characteristic distinguishes fat-soluble vitamins from passive cellular lipid components?

Options :

1. They are only involved in energy storage
2. They form structural frameworks without active roles
3. They participate in metabolic signaling and enzymatic reactions
4. They remain inactive until oxidized by enzymes

Question Number : 10 Question Id : 4410092155717 Question Type : MCQ

A researcher is designing a study on oxidative damage in cell membranes. Which vitamin deficiency should be induced to best observe increased membrane lipid peroxidation?

Options :

1. Vitamin B2 deficiency
2. Vitamin A deficiency
3. Vitamin E deficiency
4. Vitamin K deficiency

Question Number : 11 Question Id : 4410092155733 Question Type : MCQ

What role does a protein kinase play in the regulation of the branched-chain alpha-keto acid dehydrogenase complex?

Options :

1. It adds a phosphate group to inactivate the enzyme
2. It degrades the enzyme complex
3. It adds a phosphate group to activate the enzyme
4. It removes a phosphate group to activate the enzyme

Question Number : 12 Question Id : 4410092155735 Question Type : MCQ

Which regulatory mechanism is shared by both the branched-chain alpha-keto acid dehydrogenase complex and the pyruvate dehydrogenase complex?

Options :

1. Regulation by proteolytic cleavage
2. Regulation by feedback inhibition
3. Regulation by phosphorylation and dephosphorylation
4. Regulation by allosteric activation only

Question Number : 13 Question Id : 4410092155737 Question Type : MCQ

What does enzyme turnover refer to in the context of enzyme regulation?

Options :

1. Constant synthesis and degradation of enzyme
2. Activation of enzyme by cofactor
3. Allosteric regulation of enzyme activity
4. Feedback inhibition of enzyme function

Question Number : 14 Question Id : 4410092155731 Question Type : MCQ

A peptide with the C-terminal sequence Met–Trp–Ser–Pro is generated after chymotrypsin digestion. What does the presence of phenylalanine immediately before methionine indicate?

Options :

1. Phenylalanine is the C-terminal residue of the generated peptide
2. Phenylalanine marks the N-terminal of the peptide
3. Phenylalanine blocks further cleavage by chymotrypsin
4. Phenylalanine is the last residue cleaved due to chymotrypsin specificity

Question Number : 15 Question Id : 4410092155734 Question Type : MCQ

What is the effect of phosphorylation on the branched-chain alpha-keto acid dehydrogenase complex when rats have a diet low in branched-chain amino acids?

Options :

1. The enzyme complex increases its gene expression
2. The enzyme complex binds to cofactors more strongly
3. The enzyme complex becomes inactivated
4. The enzyme complex becomes activated

Question Number : 16 Question Id : 4410092155736 Question Type : MCQ

Which mechanism can modulate the catalytic activity of an enzyme molecule already present in the cell?

Options :

1. Stabilization of mRNA
2. Activity of transcription factors
3. Regulation of protein synthesis
4. Allosteric regulation or covalent alteration

Question Number : 17 Question Id : 4410092155642 Question Type : MCQ

If a biochemical reaction has a negative ΔG , what does this imply about the reaction?

Options :

1. The reaction requires a gain of free energy to proceed
2. The reaction is at equilibrium with no net change
3. The reaction moves the system further away from equilibrium
4. The reaction proceeds spontaneously with loss of free energy

Question Number : 18 Question Id : 4410092155646 Question Type : MCQ

Which pathway provides reducing equivalents for fatty acid synthesis in cells?

Options :

1. TCA cycle
2. Glycolysis
3. Pentose phosphate pathway
4. Urea cycle

Question Number : 19 Question Id : 4410092155651 Question Type : MCQ

Which molecule is produced alongside acetyl-CoA in the final round of beta-oxidation of odd-chain fatty acids?

Options :

1. Fumarate
2. Succinyl-CoA
3. Propionyl-CoA
4. Malonyl-CoA

Question Number : 20 Question Id : 4410092155655 Question Type : MCQ

Which pair correctly matches an enzyme of the fatty acid synthase complex with its function?

Options :

1. Acyl carrier protein: catalyzes reduction of beta-keto group
2. Malonyl-CoA–ACP transferase: transfers acetyl group to ACP thiol group
3. Acetyl-CoA–ACP transacetylase: transfers acetyl group to Cys thiol group
4. Acetyl-CoA–ACP transacetylase: catalyzes dehydration step

Question Number : 21 Question Id : 4410092155656 Question Type : MCQ

If a drug inhibits the cyclization of squalene 2,3-epoxide, which immediate effect would you expect in animal cells?

Options :

1. Farnesyl pyrophosphate would accumulate
2. Cholesterol synthesis would accelerate
3. Mevalonate production would increase
4. Lanosterol formation would be blocked

Question Number : 22 Question Id : 4410092155652 Question Type : MCQ

A person has a mutation that prevents malonyl-CoA from inhibiting carnitine acyltransferase I. What metabolic consequence would be expected when this person consumes a high-carbohydrate meal?

Options :

1. Phospholipid breakdown is enhanced in cytosol
2. Triacylglycerol synthesis is suppressed in cytosol
3. Fatty acid entry into mitochondria is decreased
4. Fatty acid oxidation continues despite high glucose levels

Question Number : 23 Question Id : 4410092155657 Question Type : MCQ

In the biosynthesis of phospholipids, which step involves the formation of a phosphodiester linkage?

Options :

1. Addition of the hydrophilic head group to the backbone
2. Attachment of fatty acid to the backbone
3. Synthesis of the backbone molecule
4. Exchange of the fatty acid moiety

Question Number : 24 Question Id : 4410092155661 Question Type : MCQ

A patient lacks pyridoxal phosphate. Which step in serine metabolism is affected?

Options :

1. Hydrolysis to cysteine
2. Transamination of 3-phosphoserine
3. The oxidation of 3-phosphoglycerate
4. The reversible conversion of serine to glycine

Question Number : 25 Question Id : 4410092155663 Question Type : MCQ

What is the metabolic consequence of pyrimidine degradation that contributes to urea synthesis?

Options :

1. succinyl-CoA generation
2. methylmalonyl-CoA formation
3. propionyl-CoA formation
4. ammonia production

Question Number : 26 Question Id : 4410092155665 Question Type : MCQ

A deficiency in the enzyme that produces 5-phosphoribosyl 5-pyrophosphate would impact the biosynthesis of which nucleotide coenzyme?

Options :

1. GTP
2. CTP
3. NAD⁺
4. ATP

Question Number : 27 Question Id : 4410092155676 Question Type : MCQ

What is the role of histone proteins in chromatin structure?

Options :

1. Packaging and ordering DNA
2. Replicating DNA
3. Transporting proteins
4. Synthesizing RNA

Question Number : 28 Question Id : 4410092155681 Question Type : MCQ

Which environmental stress can *Escherichia coli* survive by modulating cell growth and division?

Options :

1. Acid shock
2. Extreme heat
3. Metal toxicity
4. Light exposure

Question Number : 29 Question Id : 4410092155683 Question Type : MCQ

What is the main reason the Miles & Misra method uses fewer plates compared to traditional spread plates when counting pure cultures?

Options :

1. Multiple drops can be placed on one plate
2. Each drop contains more bacteria
3. Plates are incubated at higher temperatures
4. Each plate uses smaller volumes

Question Number : 30 Question Id : 4410092155679 Question Type : MCQ

A mutation causes a chromatin region to remain less condensed but become resistant to nuclease-mediated degradation. What is the direct consequence for gene expression in this region?

Options :

1. Genes in this region will undergo increased methylation
2. Genes in this region are likely to be highly expressed
3. Genes in this region will replicate earlier
4. Genes in this region are likely to be transcriptionally inactive

Question Number : 31 Question Id : 4410092155680 Question Type : MCQ

In an experiment, a culture is observed to have a constant rate of cell division before declining. What phase is this and what is its key feature?

Options :

1. Death phase with declining cell numbers
2. Lag phase with cell adjustment
3. Stationary phase with equilibrium
4. Exponential phase with constant division rate

Question Number : 32 Question Id : 4410092155682 Question Type : MCQ

When isolating microorganisms from a mixed sample, which property of *Micrococcus luteus* aids its identification?

Options :

1. It forms heat-resistant spores
2. It exhibits active motility
3. It produces yellow colonies on nutrient agar
4. It is highly heat-resistant

Question Number : 33 Question Id : 4410092155386 Question Type : MCQ

During sexduction, the F pilus is a specialized appendage responsible for which of the following tasks?

Options :

1. Synthesizes new DNA strands
2. Supports cell attachment and DNA transfer
3. Confers antibiotic resistance
4. Acts as a restriction enzyme

Question Number : 34 Question Id : 4410092155385 Question Type : MCQ

Which feature distinguishes Hfr cells from F⁺ cells in bacterial conjugation?

Options :

1. Hfr cells lack the ability to form pili
2. Hfr cells possess chromosomally integrated F factor
3. Hfr cells contain the F factor as a plasmid
4. Hfr cells do not carry any F factor

Question Number : 35 Question Id : 4410092155388 Question Type : MCQ

How do intercalating agents like ethidium bromide induce mutations?

Options :

1. By directly causing base substitutions
2. By inserting between bases and inducing frameshifts
3. By breaking DNA strands at random locations
4. By removing bases from the DNA sequence

Question Number : 36 Question Id : 4410092155394 Question Type : MCQ

In insertional mutagenesis screens, why is it critical to map the exact genomic insertion site?

Options :

1. To ensure high efficiency of gene delivery
2. To identify disrupted genes and link genotype to phenotype
3. To observe resulting phenotypic changes in the organism
4. To verify the presence of the integration vector only

Question Number : 37 Question Id : 4410092155396 Question Type : MCQ

A chromosomal analysis reveals an interstitial duplication. Which cytogenetic signature indicates this structural chromosomal abnormality?

Options :

1. Segment inserted into a different chromosome
2. Duplication observed at the terminal region of the chromosome
3. Formation of a ring chromosome
4. Additional chromosomal segment within the same chromosome

Question Number : 38 Question Id : 4410092155406 Question Type : MCQ

Which of the following is a direct consequence of the phosphorylation of the C-terminal domain (CTD) of RNA polymerase II?

Options :

1. Initiation of mRNA processing events
2. Termination of transcription
3. Inhibition of mRNA synthesis
4. Degradation of synthesized RNA

Question Number : 39 Question Id : 4410092155412 Question Type : MCQ

In mammals, which base modification is most frequently observed during RNA editing?

Options :

1. Cytosine to uracil conversion
2. Adenosine to inosine conversion
3. Uracil to cytosine conversion
4. Guanine to adenine conversion

Question Number : 40 Question Id : 4410092155417 Question Type : MCQ

Which nucleotide is hydrolyzed to drive amino acid activation during tRNA aminoacylation?

Options :

1. UTP
2. ATP
3. CTP
4. GTP

Question Number : 41 Question Id : 4410092155411 Question Type : MCQ

Which conserved cis-acting sequence serves as the primary polyadenylation signal that directs 3'-end processing of most eukaryotic pre-mRNAs?

Options :

1. CA-rich element
2. GU-U rich downstream element
3. AGGAGG sequence
4. AAUAAA consensus sequence

Question Number : 42 Question Id : 4410092155415 Question Type : MCQ

Which modification of eIF2 is primarily responsible for the inhibition of global protein synthesis during cellular stress?

Options :

1. Methylation of eIF2 beta subunit
2. Acetylation of eIF2 gamma subunit
3. Ubiquitination of eIF2 alpha subunit
4. Phosphorylation of eIF2 alpha subunit

Question Number : 43 Question Id : 4410092155422 Question Type : MCQ

How does the Cro protein promote the lytic cycle in bacteriophage λ ?

Options :

1. Directly activates lytic gene transcription
2. Binds to the origin of replication
3. Represses lambda repressor gene transcription
4. Selectively degrades host mRNA

Question Number : 44 Question Id : 4410092155414 Question Type : MCQ

A mutant cell line exhibits an increased 40S:60S ribosomal subunit ratio. Which defect most likely accounts for this finding?

Options :

1. Increased eIF4E expression
2. Hyperactive aminoacyl-tRNA synthetases
3. Impaired 28S rRNA maturation
4. Enhanced methylation of 18S rRNA

Question Number : 45 Question Id : 4410092155416 Question Type : MCQ

Which event directly triggers polypeptide release during eukaryotic translation termination?

Options :

1. eRF1 recognition of the stop codon at the A site
2. GTP hydrolysis by eRF3 causing ribosome dissociation
3. Peptidyl transferase-mediated peptide bond formation
4. Ribosome recycling factor binding to the ribosome

Question Number : 46 Question Id : 4410092155418 Question Type : MCQ

Which reaction is directly catalyzed by aminoacyl-tRNA synthetase?

Options :

1. Formation of a peptide bond between amino acids
2. Formation of an ester bond between amino acid and tRNA
3. Cleavage of mRNA during splicing
4. Hydrolysis of GTP during translation

Question Number : 47 Question Id : 4410092155423 Question Type : MCQ

Which mechanism allows RNA viruses to maximize protein coding capacity from a compact genome?

Options :

1. Host-mediated mRNA splicing
2. Utilization of overlapping reading frames
3. Leaky scanning by ribosomes
4. Use of internal ribosome entry sites for translation

Question Number : 48 Question Id : 4410092155425 Question Type : MCQ

Which mechanism exemplifies epigenetic regulation of gene expression in eukaryotes?

Options :

1. Alternative splicing of pre-mRNA
2. Promoter CpG methylation
3. Point mutations in the coding sequence
4. Translation regulation by ribosome availability

Question Number : 49 Question Id : 4410092155429 Question Type : MCQ

A researcher inserts additional nucleotides between the trpL leader sequence and the structural genes, increasing the distance. Which regulatory process is most likely to be disrupted?

Options :

1. Attenuation dependent on coupled transcription and translation
2. Catabolite repression by glucose availability
3. Feedback inhibition of tryptophan biosynthesis
4. Repression by the trp repressor protein

Question Number : 50 Question Id : 4410092155483 Question Type : MCQ

Which characteristic response property is the hallmark feature that distinguishes adaptive immunity from innate immunity?

Options :

1. Immunological memory after first exposure
2. Immediate response without prior exposure
3. Non-specific recognition of pathogen patterns
4. Physical barrier protection at body surfaces

Question Number : 51 Question Id : 4410092155486 Question Type : MCQ

The high specificity of antigen-antibody binding is attributed to non-covalent forces acting within which structural region of the antibody?

Options :

1. Complementarity-determining regions of Fab
2. Constant region of heavy chain Fc
3. Hinge region between Fab and Fc
4. J chain of polymeric antibody form

Question Number : 52 Question Id : 4410092155488 Question Type : MCQ

Which molecule blocks the MHC class II binding cleft prior to the loading of processed antigens?

Options :

1. Invariant chain
2. TAP1 protein
3. Calreticulin
4. HLA-DM dimer

Question Number : 53 Question Id : 4410092155495 Question Type : MCQ

Which complement regulatory protease irreversibly inactivates both C3b and C4b by proteolytic cleavage?

Options :

1. Factor H
2. Factor I
3. Properdin
4. C1 inhibitor

Question Number : 54 Question Id : 4410092155497 Question Type : MCQ

Which inflammasome sensor protein oligomerizes after binding cytosolic double-stranded DNA?

Options :

1. RIG-I receptor
2. NLRC4 receptor
3. NLRP3 receptor
4. AIM-2 receptor

Question Number : 55 Question Id : 4410092155498 Question Type : MCQ

Which host antigen-presenting molecules specifically present mycobacterial lipid antigens to specialized T cells?

Options :

1. CD1 proteins
2. HLA class II
3. TLR2 sensors
4. MR1 proteins

Question Number : 56 Question Id : 4410092155514 Question Type : MCQ

Auxin (IAA) is degraded oxidatively in plant tissues primarily by which enzyme?

Options :

1. Amylase enzyme activity
2. Cellulase enzyme activity
3. IAA oxidase enzyme activity
4. Nitrate reductase enzyme activity

Question Number : 57 Question Id : 4410092155510 Question Type : MCQ

Ammonium produced during nitrate assimilation in plants is first incorporated into an organic compound by which enzyme?

Options :

1. Glutamine synthetase converting glutamate to glutamine
2. Aspartate aminotransferase converting OAA to aspartate
3. Glutamate dehydrogenase converting α -ketoglutarate to glutamate
4. Asparagine synthetase converting aspartate to asparagine

Question Number : 58 Question Id : 4410092155515 Question Type : MCQ

Which physiological effect of auxin is primarily responsible for apical dominance in plants?

Options :

1. Suppression of axillary bud growth
2. Stimulation of axillary bud growth
3. Promotion of leaf senescence
4. Inhibition of adventitious root formation

Question Number : 59 Question Id : 4410092155505 Question Type : MCQ

Which chloroplast enzyme catalyzes malate decarboxylation during the daytime in NADP-malic enzyme (NADP-ME) type CAM plants?

Options :

1. NADP-malic enzyme
2. PEP carboxykinase
3. ATP citrate lyase
4. RuBisCO oxygenase

Question Number : 60 Question Id : 4410092155511 Question Type : MCQ

Boron deficiency impairs plant cell wall integrity by preventing cross-linking of which specific cell wall component?

Options :

1. Extensin glycoprotein units via isodityrosine crosslinks
2. Lignin polymer units via peroxidase-catalyzed bonds
3. Cellulose microfibrils via hydrogen bond formation
4. Rhamnogalacturonan II via borate ester bonds

Question Number : 61 Question Id : 4410092155522 Question Type : MCQ

Which hormone, secreted by an empty stomach, increases food intake by acting on hypothalamic feeding centers?

Options :

1. Ghrelin
2. Leptin
3. Insulin
4. Cholecystokinin

Question Number : 62 Question Id : 4410092155532 Question Type : MCQ

Which enzyme directly converts fibrinogen to fibrin in the final common pathway of the coagulation cascade?

Options :

1. Thrombin acting on fibrinogen
2. Plasmin acting on fibrinogen
3. Factor Xa acting on fibrinogen
4. Tissue factor acting on fibrinogen

Question Number : 63 Question Id : 4410092155518 Question Type : MCQ

Which site is responsible for action potential regeneration during saltatory conduction in myelinated axons?

Options :

1. Schwann cell membrane surface
2. Dendritic spine synapse sites
3. Axon hillock trigger zone
4. Nodes of Ranvier along axon

Question Number : 64 Question Id : 4410092155529 Question Type : MCQ

Erythropoietin, the primary regulator of red blood cell production, is secreted mainly by which organ in adults?

Options :

1. Kidney
2. Thymus
3. Pancreas
4. Bone marrow

Question Number : 65 Question Id : 4410092155520 Question Type : MCQ

The binding of calcium to troponin C exposes the myosin-binding site on actin through which structural change?

Options :

1. Sarcoplasmic reticulum membrane depolarization
2. Tropomyosin shift exposing binding site
3. Actin filament shortening directly
4. Myosin head detachment from actin

Question Number : 66 Question Id : 4410092155528 Question Type : MCQ

The biconcave shape of mature erythrocytes is maintained primarily by which membrane cytoskeletal protein network?

Options :

1. Actin-myosin contractile network
2. Spectrin-ankyrin protein network
3. Microtubule-based axonemal network
4. Intermediate filament keratin network

Question Number : 67 Question Id : 4410092155531 Question Type : MCQ

Hemolytic disease of the newborn due to Rh incompatibility occurs mainly because maternal anti-D antibodies belong to which immunoglobulin class?

Options :

1. IgE bound to maternal mast cells
2. IgG crossing the placental barrier
3. IgM remaining in maternal circulation
4. IgA secreted in colostrum only

Question Number : 68 Question Id : 4410092155539 Question Type : MCQ

Fluoroquinolones inhibit bacterial DNA replication primarily by targeting which enzyme?

Options :

1. DNA gyrase
2. RNA polymerase
3. DNA ligase
4. Reverse transcriptase

Question Number : 69 Question Id : 4410092155545 Question Type : MCQ

Which type of mutation in the β -globin gene most commonly causes β -thalassemia?

Options :

1. Point mutations affecting splicing
2. Large deletions of whole gene
3. Trinucleotide repeat expansion
4. Chromosomal translocation only

Question Number : 70 Question Id : 4410092155554 Question Type : MCQ

The intrinsic pathway of apoptosis is triggered by release of which mitochondrial protein into the cytosol?

Options :

1. Cytochrome c
2. Caspase-8 enzyme
3. FADD adaptor
4. TNF-alpha ligand

Question Number : 71 Question Id : 4410092155537 Question Type : MCQ

Which structure does vancomycin primarily bind to in order to inhibit bacterial cell wall synthesis?

Options :

1. D-Ala-D-Ala terminal peptide
2. Penicillin-binding protein site
3. Peptidoglycan glycan backbone
4. Beta-lactamase active enzyme site

Question Number : 72 Question Id : 4410092155549 Question Type : MCQ

The Philadelphia chromosome generates the BCR-ABL fusion protein, which drives chronic myeloid leukemia (CML) through constitutive activation of _____.

Options :

1. Tyrosine kinase activity
2. BAX activity
3. GTPase enzyme activity
4. Phosphatase enzyme activity

Question Number : 73 Question Id : 4410092155550 Question Type : MCQ

A RAS protein is locked in its GTP-bound active state due to impaired GTPase activity. Which genetic alteration is most likely responsible?

Options :

1. Point mutation impairing GTPase activity of RAS
2. Deletion of the farnesylation site of RAS only
3. Point mutation increasing the binding affinity of RAS for GAP
4. Loss-of-function mutation only in the Sos guanine nucleotide exchange factor

Question Number : 74 Question Id : 4410092155577 Question Type : MCQ

Why is the skin generally a less efficient exposure route for most inorganic heavy metals compared to the lungs?

Options :

1. Metals are rapidly metabolized only in the skin
2. Keratin-rich outer skin layer limits diffusion
3. Skin lacks blood flow for systemic distribution
4. Heavy metals require only active transport through skin

Question Number : 75 Question Id : 4410092155580 Question Type : MCQ

Why is a skin rash resulting from a pesticide splash classified as local toxicity, whereas nausea caused by the same pesticide exposure is considered systemic toxicity?

Options :

1. The rash occurs only after the chemical is metabolized.
2. The rash is restricted to the point of chemical contact.
3. Nausea is a result of direct irritation to the epidermis.
4. Skin rashes only occur through the inhalation route.

Question Number : 76 Question Id : 4410092155582 Question Type : MCQ

Which hepatic enzyme system is primarily responsible for the oxidation of xenobiotics during Phase I biotransformation?

Options :

1. Transferases
2. Cytochrome P450 enzymes
3. Acetylcholinesterase
4. Succinate dehydrogenase

Question Number : 77 Question Id : 4410092155584 Question Type : MCQ

DNA ligase catalyzes the formation of which bond between adjacent DNA fragments?

Options :

1. Glycosidic
2. Hydrogen
3. Phosphodiester
4. Peptide

Question Number : 78 Question Id : 4410092155587 Question Type : MCQ

Why is a linearized plasmid vector treated with alkaline phosphatase before ligation with an insert?

Options :

1. To stimulate T4 DNA ligase activity
2. To prevent vector self-ligation
3. To ensure only correct insert orientation
4. To protect the multiple cloning site from exonuclease digestion

Question Number : 79 Question Id : 4410092155603 Question Type : MCQ

Why is formaldehyde added to the gel during Northern blotting?

Options :

1. To maintain the pH of the gel
2. To prevent RNA from forming secondary structures
3. To enhance ethidium bromide staining
4. To increase the migration rate of RNA only

Question Number : 80 Question Id : 4410092155601 Question Type : MCQ

In a standard blue-white screening system, a white bacterial colony on X-gal-containing selective medium most likely indicates _____.

Options :

1. Successful insertional inactivation of lacZ by foreign DNA
2. Absence of the plasmid vector
3. Loss of antibiotic resistance gene
4. Failure of transformation

Question Number : 81 Question Id : 4410092155605 Question Type : MCQ

In cancer research, RNAi-based therapy is being explored to treat malignancies characterized by the over-expression of which specific oncogene?

Options :

1. CFTR
2. Factor VIII
3. HPRT
4. MYC

Question Number : 82 Question Id : 4410092155611 Question Type : MCQ

Recombinant DNA technology enables diagnostic ELISA kits to use pathogen-specific _____ as a safer alternative to live cultured virus.

Options :

1. Restriction enzymes
2. Recombinant antigens
3. Selectable markers
4. Origin of replication sequences

Question Number : 83 Question Id : 4410092155585 Question Type : MCQ

In recombinant DNA technology, which enzyme catalyzes the covalent joining of DNA fragments derived from different sources?

Options :

1. DNA polymerase I
2. DNA ligase
3. Alkaline phosphatase
4. DNA helicase

Question Number : 84 Question Id : 4410092155588 Question Type : MCQ

Two bacteriophages are used to infect separate E. coli cultures: one contains radiolabeled DNA (culture A), while the other contains a radiolabeled protein coat (culture B). After infection, the bacteria are separated from the phage particles and tested for radioactivity. Which culture shows radioactivity?

Options :

1. Both cultures
2. Neither culture
3. Culture A
4. Culture B

Question Number : 85 Question Id : 4410092155594 Question Type : MCQ

What is a major advantage of using yeast or mammalian cells over E. coli for producing human therapeutic proteins?

Options :

1. Faster cell division rate
2. Lower culture medium cost
3. Ability to perform post-translational modifications
4. Absence of a nuclear membrane

Question Number : 86 Question Id : 4410092155595 Question Type : MCQ

Which of the following features makes fungal cell systems useful for expressing many eukaryotic recombinant proteins?

Options :

1. They lack cellular mechanisms for degrading recombinant proteins
2. They produce proteins exclusively within the nucleus
3. They eliminate the need for recombinant DNA vectors
4. They can perform protein folding and several post-translational modifications

Question Number : 87 Question Id : 4410092155607 Question Type : MCQ

Match the following:

List I	List II
I. RT-PCR	1. Quantification of DNA amplification in real time
II. qPCR	2. Amplification of RNA via cDNA intermediate
III. RAPD-PCR	3. DNA fingerprinting and genetic variation analysis
IV. ASO-PCR	4. Detection of specific point mutations

Options :

1. I-2, II-1, III-3, IV-4
2. I-1, II-2, III-3, IV-4
3. I-2, II-3, III-1, IV-4
4. I-3, II-1, III-2, IV-4

Question Number : 88 Question Id : 4410092155614 Question Type : MCQ

Which molecular technique is most commonly used for bacterial strain identification based on DNA fragment patterns?

Options :

1. ELISA
2. Gram staining
3. RFLP
4. Catalase test

Question Number : 89 Question Id : 4410092155625 Question Type : MCQ

In the production of polyclonal antibodies, which avian immunoglobulin is harvested from chicken egg yolks as an alternative to mammalian serum IgG?

Options :

1. IgY
2. IgM
3. IgE
4. IgA

Question Number : 90 Question Id : 4410092155630 Question Type : MCQ

Which type of radioactive decay involves the conversion of a neutron into a proton with the simultaneous emission of a light, negatively charged particle?

Options :

1. Alpha (α) emission
2. Positron (β^+) emission
3. Negatron (β^-) emission
4. Electron capture

Question Number : 91 Question Id : 4410092155617 Question Type : MCQ

Arrange the following steps of a typical protein purification protocol in the CORRECT sequential order.

- (i) Ion-exchange chromatography
- (ii) Cell lysis
- (iii) Ammonium sulfate precipitation
- (iv) Gel filtration (polishing)

Options :

1. (ii) $\rightarrow$ (iii) $\rightarrow$ (i) $\rightarrow$ (iv)
2. (i) $\rightarrow$ (ii) $\rightarrow$ (iii) $\rightarrow$ (iv)
3. (iii) $\rightarrow$ (ii) $\rightarrow$ (i) $\rightarrow$ (iv)
4. (ii) $\rightarrow$ (i) $\rightarrow$ (iii) $\rightarrow$ (iv)

Question Number : 92 Question Id : 4410092155619 Question Type : MCQ

Choose the CORRECT answer based on the statements below.

Assertion (A): Alkaline phosphatase treatment prevents vector self-ligation.

Reason (R): It removes the 5'-phosphate groups required for DNA ligase activity.

Options :

1. Both A and R are true, and R correctly explains A.
2. Both A and R are true, but R does not explain A.
3. A is true, but R is false.
4. A is false, but R is true.

Question Number : 93 Question Id : 4410092155621 Question Type : MCQ

Choose the CORRECT answer based on the statements below.

Assertion (A): Oligo deoxythymidine primers are used to initiate first-strand cDNA synthesis in eukaryotes.

Reason (R): Eukaryotic mRNA has a 3' polyadenylate tail complementary to oligo deoxythymidine.

Options :

1. Both A and R are true, and R correctly explains A.
2. Both A and R are true, but R does not explain A.
3. A is true, but R is false.
4. A is false, but R is true.

Question Number : 94 Question Id : 4410092155629 Question Type : MCQ

In hypothesis testing, what error is committed when a true null hypothesis is rejected?

Options :

1. Type I error
2. Type II error
3. Standard error
4. Sampling error

Question Number : 95 Question Id : 4410092155616 Question Type : MCQ

Match the following

List I	List II
1. Lysozyme	A. Solubilisation of membranes
2. SDS	B. Digestion of bacterial cell walls
3. Proteinase K	C. Precipitation of DNA
4. Ethanol	D. Degradation of proteins

Options :

- 1-B, 2-A, 3-D, 4-C
- 1-A, 2-D, 3-B, 4-C
- 1-B, 2-C, 3-A, 4-D
- 1-D, 2-A, 3-C, 4-B

Question Number : 96 Question Id : 4410092155618 Question Type : MCQ

Match the following.

Column I	Column II
1. Ampholytes	A. Formation of pH gradient
2. Urea	B. Concentration of sample before separation
3. Stacking Gel	C. Maintain protein denaturation and solubility

Options :

- 1-A, 2-C, 3-B
- 1-B, 2-A, 3-C
- 1-C, 2-B, 3-A
- 1-A, 2-B, 3-C

Question Number : 97 Question Id : 4410092155622 Question Type : MCQ

The most suitable technique for detecting chromosomal rearrangements such as translocations, deletions, and duplications in cancer cells is _____.

Options :

1. RAPD
2. Multi-colour FISH
3. Microarray
4. Flow cytometry

Question Number : 98 Question Id : 4410092155623 Question Type : MCQ

Why does incorporation of dideoxynucleotide (ddNTP) terminate DNA synthesis in Sanger sequencing?

Options :

1. It lacks a 5'-phosphate group.
2. It lacks a 3'-OH group.
3. It blocks RNA polymerase by steric hindrance.
4. It cannot base-pair with the template.

Question Number : 99 Question Id : 4410092155624 Question Type : MCQ

Which of the following CORRECTLY matches a protein analysis technique with its primary application?

Options :

1. Northern blotting — Detection of a specific protein
2. Immunoprecipitation — Isolation of a specific protein using antibodies
3. PCR — Identification of only protein–protein interactions
4. RT-PCR — Purification of a specific protein

Question Number : 100 Question Id : 4410092155628 Question Type : MCQ

The mode of the dataset 3, 5, 5, 6, 7, 5, 8 is _____.

Options :

1. 3
2. 5
3. 6
4. 7